ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

PROJECT MANAGEMENT

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Project Management paper 1 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Answers must relate to the case study where asked. Accept answers in the student's own words and other correct examples. Use consistent accuracy in calculations.

SECTION A

QUESTION 1

1.1 1.1.1 A ✓ 1.1.2 D ✓ 1.1.3 B ✓ 1.1.4 D ✓ 1.1.5 B ✓ 1.1.6 C ✓ 1.1.7 A ✓ 1.1.8 B ✓ 1.1.9 D ✓ 1.1.10 A ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 I ✓ 1.3.2 C ✓ 1.3.3 J ✓ 1.3.4 G ✓ 1.3.5 H ✓ 1.3.6 A ✓ 1.3.7 E ✓ 1.3.8 F ✓ 1.3.9 D ✓ 1.3.10 B ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Phase named ✓ + case-related activity ✓ for FOUR: initiation – agree the goals with the donor and school, write the project charter; planning – draw up the schedule, budget, WBS and risk plan; execution – do the electrical work, install computers and network, train teachers; monitoring and controlling – track progress against the schedule and budget, handle changes (accept as part of execution); closure – hand over the lab, final report to the donor, celebrate (8)

2.2 E.g. "Install and test 20 networked computers with software in the Sizanani lab and train two teachers, within R250 000, before the third term starts in 10 weeks." ✓ specific/measurable ✓ budget/achievable ✓ time-bound (3)

2.3 Any FOUR with interest ✓✓✓✓: donor company – money well spent, publicity; principal / school – a working lab on time; teachers – trained, able to use it; learners – access to computers; parents / governing body; project team – successful project and experience; suppliers – sales (4)

[15]

QUESTION 3

3.1 Level 0: Computer lab project ✓; four relevant level-1 packages ✓✓ (e.g. room preparation, procurement, installation, training / hand-over) each with two sensible level-2 tasks ✓✓✓✓ (e.g. electrical points, burglar bars; order computers, buy furniture; cabling, install software; train teachers, hand-over ceremony); correct tree structure ✓ (8)

3.2.1 Start → A ✓; A → B and A → C ✓; B → D and C → D ✓; D → E and D → F ✓; E → G and F → G ✓; G → End, nodes labelled with durations ✓ (6)

3.2.2 A–B–D–E–G = 2 + 4 + 6 + 2 + 1 = 15 days ✓; A–B–D–F–G = 16 days ✓; A–C–D–E–G = 14 days; A–C–D–F–G = 15 days ✓ (all four for full marks) (3)

3.2.3 A–B–D–F–G ✓; 16 days ✓ (2)

3.2.4 C: 4 − 3 = 1 day ✓ (B takes 4 days, C 3 days); E: 3 − 2 = 1 day ✓ (F takes 3, E 2) (2)

3.2.5 Bars on a day scale: A 0–2, B 2–6, C 2–5, D 6–12, E 12–14, F 12–15, G 15–16 ✓✓✓; labelled axes / activity names ✓ (4)

[25]

QUESTION 4

4.1 $\text{Computers } 20 \times \text{R}8\,500 = \text{R}170\,000$ ✓; $\text{total} = 170\,000 + 25\,000 + 18\,000 + 12\,000 + 6\,000 + 4\,000 = \text{R}235\,000$ ✓✓; $\text{contingency} = 250\,000 - 235\,000 = \text{R}15\,000$ ✓ (4)

4.2 Any FOUR risks ✓ with mitigation ✓: late delivery of equipment – order early, use reliable suppliers; theft of computers – burglar bars, alarm, insurance, security guard; power failures / load-shedding – UPS, surge protection; costs higher than budget – contingency fund, get quotes; teachers not available for training – schedule early, after-school sessions; bad weather delaying work (8)

4.3 A document (table) that lists all identified risks and how they will be managed ✓. Any TWO columns ✓✓: risk description; likelihood (probability); impact; risk rating; mitigation / response; owner (person responsible); status (3)

[15]

QUESTION 5

5.1 Any THREE: weekly progress reports by e-mail ✓; regular meetings with the principal ✓; photographs / updates on a WhatsApp group ✓; a project dashboard / Gantt chart showing progress; site visits for the donor (3)

5.2 Any FOUR: good leader / motivator ✓; good communicator ✓; organised / good planner ✓; problem solver / decision maker ✓; honest; manages time well; handles conflict; understands the technology (4)

5.3 It is a change to the scope (scope creep) ✓; it must go through change control: write a change request ✓, assess the effect on time, cost and risk ✓, and get approval from the sponsor / donor before doing it, or plan it as a separate project ✓. (4)

5.4 Any FOUR: hand over the lab and get formal sign-off / acceptance ✓; hand over documentation (network diagram, passwords, warranties) ✓; final financial report to the donor ✓; record lessons learnt ✓; release the team and resources; celebrate success; archive project records (4)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE